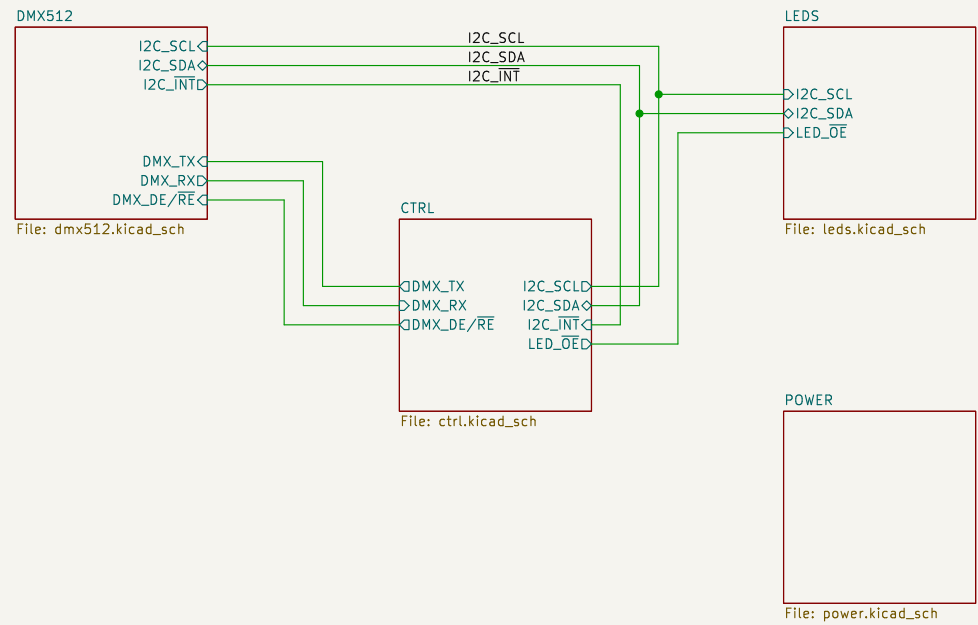


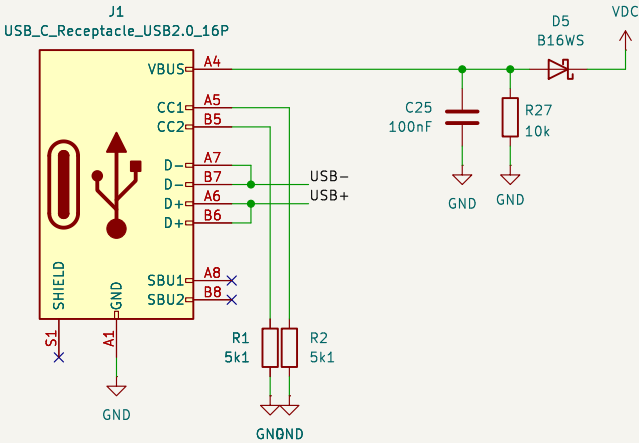
LED Beacon



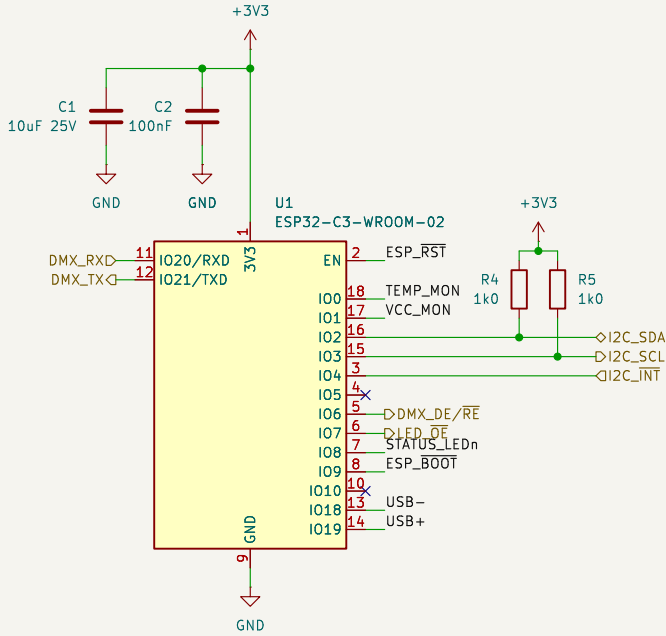
- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title: LED Beacon		
Size: A4	Date: 2026-06-05	Rev: 1
KiCad E.D.A. 9.0.9		Id: 1/5

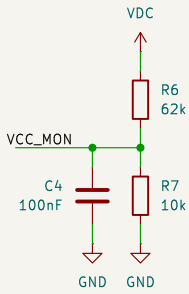
USB Programming



Controller

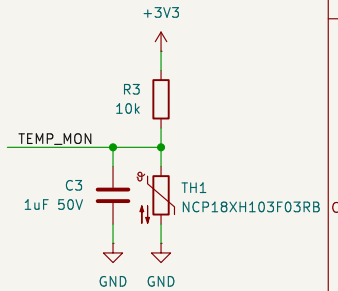


VCC Monitoring

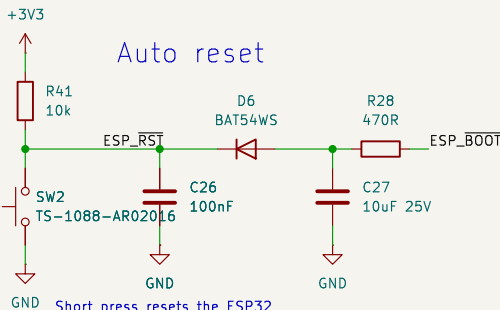


Voltage Monitor GND GND
Vmax is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) \div 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
1lsb = 4,271μV

Temperature Monitoring

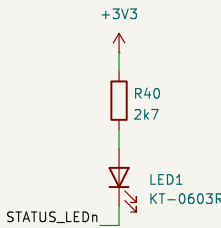


Auto reset



Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LED



IO8 needs to be held high to boot
so LED & resistor are high side.
Set STATUS_LEDn to 0V to light LED.

Wires & Bits

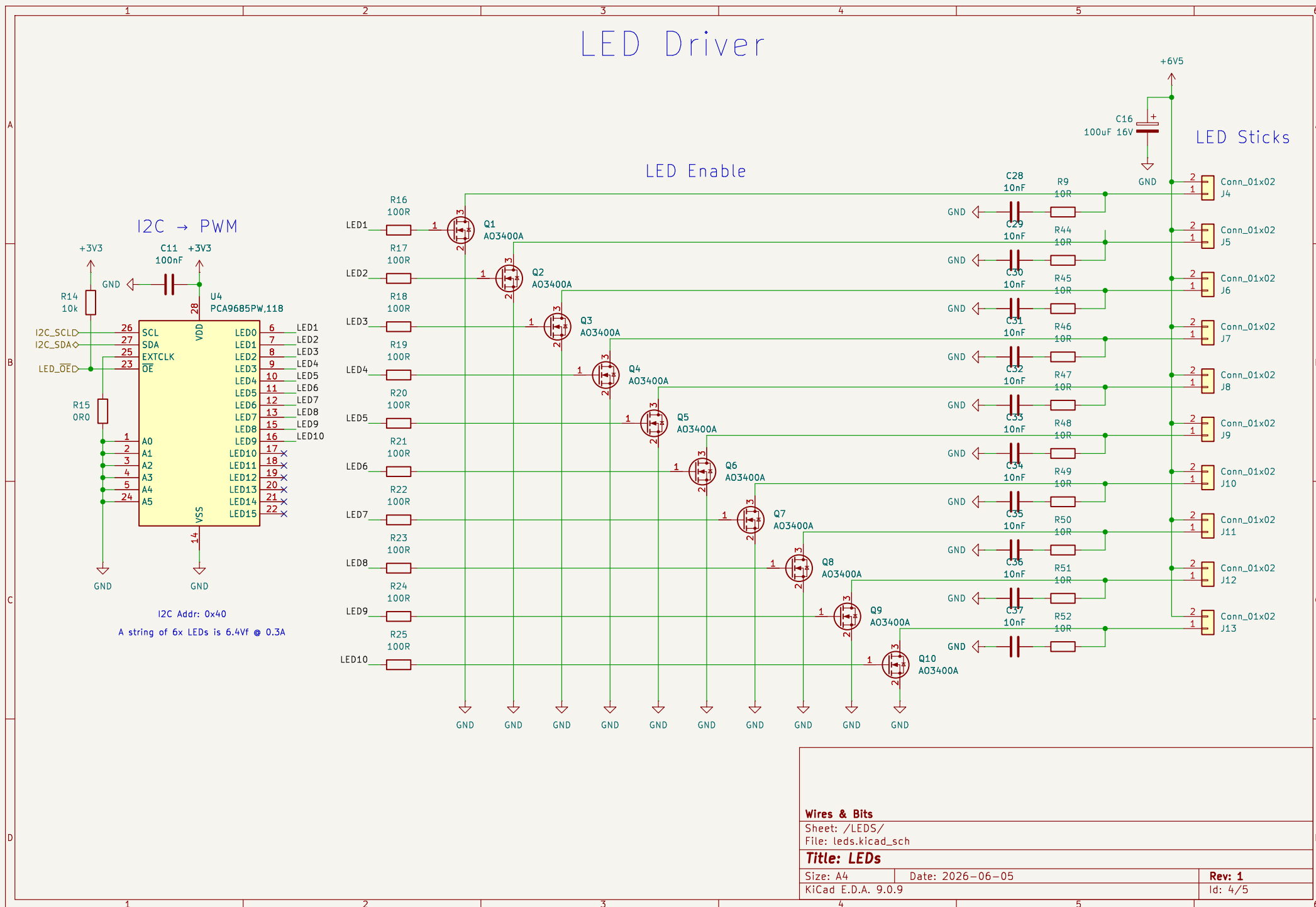
Sheet: /CTRL/
File: ctrl.kicad_sch

Title: Controller

Size: A4	Date: 2026-06-05
KiCad E.D.A. 9.0.9	

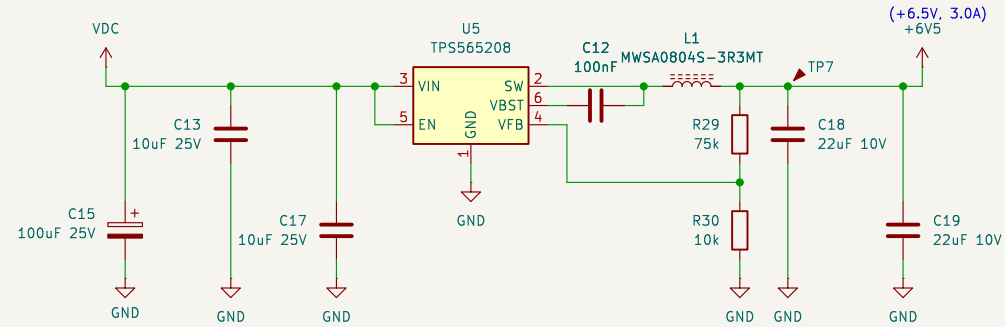
Rev: 1
Id: 2/5

LED Driver

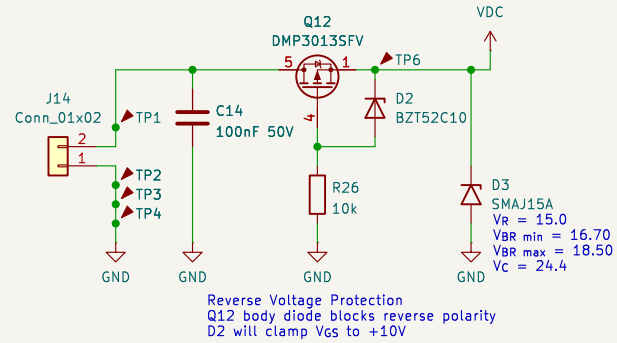
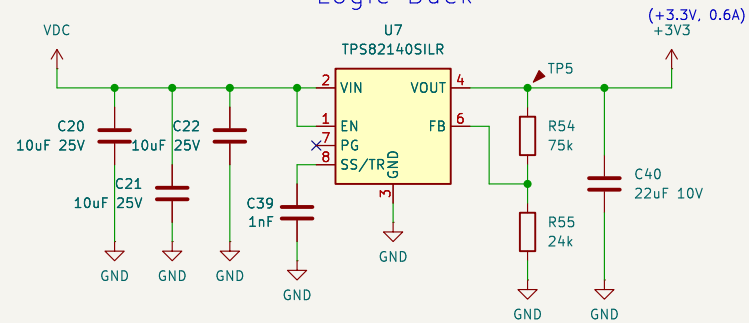


Power Supplies

LED Buck



Logic Buck



EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title: Power

Size: A4 Date: 2026-06-05
KiCad E.D.A. 9.0.9

Rev: 1
Id: 5/5